Learning Cyrillic Alphabet in Context: importance of letter position through eye-tracking

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Abstract

The position of letters has been shown to play an important role in word processing. This study explores whether such positional differences can be leveraged to facilitate the acquisition of new alphabets. To achieve this, we designed an experiment in which Cyrillic letters substituted their Latin counterparts within English sentences. Participants were instructed to read these sentences while collected eye-tracking data. Further, we directly assessed their learning outcomes. Our findings indicate that learning letters in this manner is generally achievable, with the highest accuracy observed when replacing the first letter in a word. Replacing letters in other positions showed no significant impact on learning outcomes.

1 Introduction

Learning a language offers numerous cognitive benefits. Research indicates that bilingualism enhances cognitive abilities, metalinguistic awareness, and provides protection against age-related cognitive decline ([12], [3]). Neuroscience studies further reveal that proficiency in multiple languages induces structural changes in the brain, particularly in regions associated with language processing and cognitive control ([9]). [8]

Learning a language with a different writing system can be particularly challenging. Before being able to read or write, it is necessary to understand how each new symbol is mapped to a specific sound. The cognitive workload required to master this involves complex psycholinguistic processes of mapping *graphemes*, the smallest meaningful unit in a writing system, to *phonemes*, the smallest distinct unit of sound in a spoken language [10]. This process involves different cognitive mechanisms, such as visual perception, attention, phonological awareness, and working memory [2, 4, 5], . Learners must establish a robust mental representation of new symbols and their corresponding sounds, simultaneously developing decoding of symbols and linguistic comprehension [6].

Research in visual word recognition has revealed the crucial role of letter position in word processing. [7] remarks how the first letter of a word has a privileged role over interior letters, no matter the lateral interference, while the last letter can have a higher importance if the interference from the other letters is lower. The authors of [11] and [1] expand on this topic exploring whether letters are processed in parallel, reporting once again a clear positional advantage of the first letter.

We aim to investigate whether these positional advantages can be employed to facilitate the learning of new writing systems. We test this on the Cyrillic alphabet by systematically replacing certain Latin characters with their Cyrillic counterparts in contextualised sentences, capturing eye-tracking data of participants reading them. We hypothesize that learners would better acquire the letters replaced at the beginning or end of a word than those replaced in the middle.

2 Methods

The experiment consisted of two phases: a learning phase, where participants viewed English sentences with embedded Cyrillic letters, and a testing phase, where their ability to translate the Cyrillic letters was assessed.

2.1 Participants

Twenty naive observers participated in the experiment, of which 12 were women and 8 were men. All observers had normal or corrected-to-normal vision. All observers gave written informed consent to participate in the study.

2.2 Stimuli and Apparatus

The stimuli in the learning phase consisted of English sentences in which one letter was replaced by its Cyrillic counterpart. We varied the positions of these replacements. A letter can be replaced at the start, end, or middle of a word. For each, of the replacement positions, three different letters were used. These letters were selected to reflect similar difficulty across all positions. Therefore, each position had a letter that was similar to a letter in the Greek alphabet, one unique to Cyrillic, and one that is a digraph. An overview of the English Cyrillic counterparts is given in Table 1.

Start	Middle	End
б - b	л - l	д - d
ш - sh	ц - ts	ч - ch
г - g	п - p	ф - f

Table 1: Cyrillic letters and Latin counterparts used, grouped by their position.

We used a second set of stimuli to probe for participants learning. For each letter used, they were asked once to give the corresponding translation in English. Participants could choose between three different possible answers, one of which was correct. An example of both learning and testing trials is given in Figure 1.

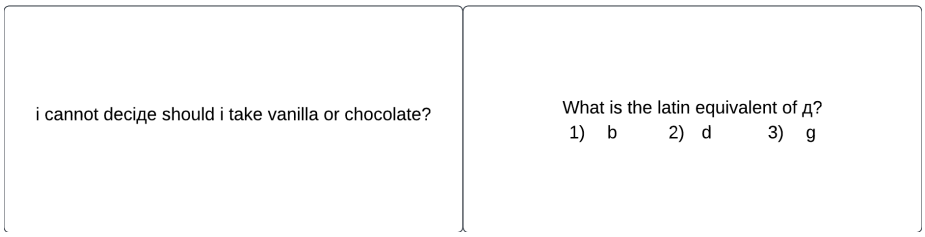


Figure 1: An example of a learning trial (left) and a testing trial (right).

Eye-tracking data was collected using a Tobii Pro eyetracker, operating at a sampling rate of 1200 Hz. The eyetracker was mounted under a desktop monitor and calibrated before each session using a 9-point calibration procedure to ensure accuracy. The system recorded participants gaze data, including fixation points, saccades, and pupil diameter.

Stimuli were displayed on a 24-inch LCD monitor with a resolution of 1920 $\times$ 1080 pixels and a refresh rate of 60 Hz. Participants were seated at an appropriate distance from the screen, and they were instructed to move as little as possible.

Data was recorded using Tobii Pro Lab software on a Windows PC. All experiments were conducted with participants alone in the room to minimize distractions.

2.3 Procedure

Our goal was to see if Cyrillic alphabet can be learned by switching out letters in sentences and how this is affected by letter positioning. During the learning phase, observers saw a sentence displayed in the centre of the screen with one of the letters replaced by its Cyrillic counterpart in one trial. Observers were instructed to read the sentence out loud and then continue to the next stimulus, by pressing the space bar. During the testing phase, one trial consisted of one question where a Cyrillic letter had to be translated into the Latin alphabet. Three options were given, of which only one was correct. The observer could answer by pressing one, two, or three on the keyboard.

2.4 Experimental Design

The independent variables were the letter replacement position, which could take three values (start, middle, and end), and the replacement Cyrillic letter. The dependent variables we measured are the eye-tracking data and the outcomes of the learning trials. The stimuli were displayed in a random order. Overall, the experiment consisted of 90 learning trials, 3 positions, 3 letters each, 10 sentences for each letter, and 9 testing trials, one for each letter.

2.5 Analysis

Learning success was assessed by analysing both eye-tracking data and learning trial outcomes. We expected that successful learning would be reflected in shorter fixation durations and a reduced number of fixations on Cyrillic letters over time, indicating increased familiarity and processing efficiency. To investigate changes in processing efficiency over time, we compared eye-tracking measures (fixation duration and number of fixations) between the initial five and the final five presentations of each Cyrillic letter. Paired t-tests were used to assess the statistical significance of any observed decreases in these measures. We assumed a significance level of 0.05.

We evaluated the outcomes of the testing trials by comparing the means of the outcomes across the different starting positions using an ANOVA test; we assumed a significance level of 0.05. To analyze which group-means are different from each other we then used Tukey's HSD posthoc test with a Family-Wise Error Rate of 0.05.

3 Results

3.1 Eye Tracking

Table 2 presents the results of paired comparisons examining changes in maximum fixation duration and number of fixations between the first and last five sentences. No clear pattern of decreasing values was observed for either feature;

mean values showed both increases and decreases. Furthermore, none of the comparisons were statistically significant.

	A			B		
Letter	**Mean 1**	**Mean 2**	**p-value**	**Mean 1**	**Mean 2**	**p-value**
b	88.36	69.20	0.304	0.35	0.28	0.273
ch	262.30	215.52	0.098	1.32	1.37	0.816
d	178.54	188.03	0.724	0.98	0.88	0.520
f	232.85	259.93	0.522	1.13	1.41	0.279
ts	189.83	193.24	0.919	0.77	0.71	0.690
l	171.77	152.24	0.613	0.92	0.58	0.193
p	183.76	176.98	0.849	0.88	0.86	0.891
sh	166.56	120.94	0.227	0.76	0.46	0.195
g	128.64	118.47	0.742	0.62	0.54	0.650

Table 2: Comparison of results for maximum fixation duration (in ms) (A) and number of fixations (B) between the first and last five sentences.

3.2 Testing Trials

Table 3 shows the aggregated metrics of participants' performance, i.e. sum, mean, and standard deviation, grouped by letter position. We highlight the differences in results for letters swapped at the start of a word, at the end, or in the middle of it.

position	letter	sum	mean	std
Start	B	20	1	0
	SH	18	0.9	0.307
	G	14	0.7	0.47
End	L	8	0.4	0.502
	TS	10	0.5	0.512
	P	9	0.45	0.51
Middle	D	12	0.666	0.485
	CH	7	0.35	0.489
	F	15	0.75	0.444

Table 3: Aggregated metrics grouped by letter-position

Table 4 shows the data resulting from all of the testing trials. It shows which letters participants could recognise and correctly associate with their Latin counterparts. The rows are ordered by performance, in a descending manner.

Finally, Table 5 presents the p-values computed using an ANOVA test between the grouped categories, followed by Tukey's HSD pair-wise tests.

subj	B	CH	D	F	G	L	P	SH	TS	tot
se	1	1	1	1	1	1	1	1	1	9
ye	1	1	1	1	1	1	1	1	1	9
yu	1	1	1	1	1	1	1	1	1	9
kl	1	0	1	1	1	1	1	1	1	8
an	1	1	0	1	1	0	1	1	1	8
ol	1	0	1	1	1	0	1	1	1	7
ch	1	0	1	1	1	0	1	1	1	7
al	1	0	1	1	1	1	0	1	1	7
sa	1	0	0	1	1	1	1	1	0	6
ma	1	1	1	1	0	0	0	1	1	6
ja	1	1	0	1	0	0	1	0	0	5
re	1	0	1	1	1	0	0	1	0	5
em	1	0	1	1	1	0	0	1	0	5
lo	1	0	0	1	1	1	0	1	0	5
fr	1	0	0	1	0	1	0	1	0	4
da	1	0	1	0	1	0	0	1	0	4
di	1	1	0	0	0	0	0	1	1	4
id	1	0	0	0	1	0	0	1	0	3
am	1	0	1	0	0	0	0	0	0	2
as	1	0	0	0	0	0	0	1	0	2

Table 4: Results from testing trials

ANOVA p-value		$3.689 \cdot 10^{-6}$		
Tukey's HSD results		end vs middle	end vs start	middle vs start
	p - adj	0.339	0.000	0.001

Table 5: ANOVA and Tukey's HSD for letter position distributions

4 Discussion

We investigated whether the positional advantages observed in word processing could be leveraged to facilitate new alphabet acquisition. Participants read contextualized sentences in which specific Latin letters were systematically replaced with their Cyrillic counterparts. We collected eye-tracking data and used direct learning assessments to evaluate the impact of letter position on learning.

The results from the eye-tracking suggested that no learning happened. Neither the maximum fixation duration nor the number of fixations consistently decreased. Additionally, the computed statistical tests were not significant.

This is in contrast to the results we got for the testing trials, where clear signs of learning were found. In the Appendix we further analyse the eye-tracking data to see whether we can predict the outcomes from the testing trials. These results indicate that the eye-tracking data we collected may not be usable to

evaluate learning performance, as it was not possible to predict the outcomes better than the majority classifier accuracy.

The testing trials results show that some learning occurred. Table 4 provides insights about the general performance across participants. Exactly half of them correctly identified strictly more than five letters. Three observers scored 9 correct answers, the maximum possible. The lowest score, 3, was also reached by three participants. Overall, it seems like the learning performance is quite uniformly distributed.

A much more detailed analysis can be done by examining Table 3, where metrics are grouped by letter position. Letters swapped at the beginning of words have a clear advantage over both other two groups. Participants seem to remember them correctly on average 86.67% times, with a surprising result from the letter б, recognised every time, by every participant, and letter ш, for which the accuracy is 90%. The average accuracy is almost twice that of letters swapped at the end, which is 45%. This is in line with the findings from [7], which showed that, however the letter on the last position has some relevance when not influenced by the laterals, the first position has a much larger advantage. Surprisingly, swapped letters in the middle of a word have a higher average accuracy than those in the end; however, this difference is not significant.

Our statistical analysis, summarised in Table 5 reveals that the only significant differences are found between start and end, and start and middle.

Overall, all letters except for ч have a higher-than-chance accuracy, which is 33% since there were three possible choices to answer the questions. This indicates that participants have learnt the Cyrillic letters successfully.

Generally, learning by replacing letters seems possible, especially if letters are replaced at the start of a word. However, replacing letters at the end of a word does not seem advantageous to replacing letters in the middle of a word.

There were many possible target alphabet candidates for our study, but Cyrillic was an interesting choice. There are more speakers of languages using Cyrillic than other alternatives we considered. Other scripts, such as Georgian, Armenian, and Mongolian, were among our options. Their glyphs have very different shapes than Latin, Greek, or Cyrillic scripts. However, Cyrillic-based languages have many more speakers and learners around the world. Russian[1] alone has 225 million people speaking it either as a mother tongue or second language, while Georgian[2] and Armenian[3] only makeup 9 million, making it a more interesting choice for the potential future developments.

Confounding Factors

In our study, we did not adequately address some potential confounding factors that could influence our design and results. Firstly, some words may be more complex, making a substituted letter more challenging to remember. The

[1] Russian
[2] Georgian
[3] Armenian

same goes for letters, as certain glyphs might be more difficult to distinguish. In Cyrillic, some specific sounds are denoted by a single letter, when English uses multiple consonants, which can be perplexing. This discrepancy requires cognitive effort to adapt, as one letter in Cyrillic represents a sound that would typically involve two letters in English. Consequently, some sounds, like *ts*, are infrequent in English, appearing mainly at the end of words like 'cats', and do not fully overlap with the *ts* sound found in Slavic languages like Russian, which Cyrillic accurately depicts. Further, some letters are similar to letters in the Greek alphabet and thus potentially easier to recognize. In Appendix A.2 we analysed whether diphthongs or similarity to Greek influence the difficulty of learning a letter. Our results indicate that such an influence is not present. Additionally, letter shapes might have an impact: the least recognised letter, ч, has a shape similar to the Latin y. At the same time, the most recognised one, б, might somehow closely resemble a Latin b.

Finally, the position of the letter in the sentence may be a confounding factor as well. It may be easier to infer the sound of a letter at the end of a sentence compared to the beginning, because of the additional context.

We hope that using a sample of ten sentences for each letter will help minimize the variance of these influencing factors to an acceptable level.

Future Work

This study represents a foundational step in exploring second language acquisition, particularly with respect to learning new writing systems. A clear path for future research is the investigation of alternative scripts. While focusing on a widely used system like the Cyrillic alphabet offers practical advantages, such as broader applicability in foreign language teaching, investigating other writing systems would allow us to determine whether the effects we observed here are generalizable. This could include testing with scripts that are not inherently alphabetical, such as Arabic, Devanagari, or logographic systems like Chinese.

Long-term learning is another critical area for further exploration. Evaluating retention over extended periods would provide insight about the durability of learning gains and the effectiveness of contextual substitution methods over time.

Moreover, the Cyrillic alphabet itself offers intriguing aspects we did not explore. As an example, certain letters derived from Latin and Greek scripts have differing phonemic representations across languages. The character *H* exemplifies this phenomenon: in Russian (Cyrillic), it represents the IPA sound /*n*/, in Greek /ε : /, and in English (Latin script) /*h*/. These phonological differences, despite graphemic similarity, could represent a challenge and interference for learners accustomed to other alphabets. Exploring these factors further would enrich our understanding of the complexities involved in learning non-native scripts.

4.1 Conclusion

Research in visual word recognition has revealed the crucial role of letter position in word processing. We investigated whether these positional differences can be employed to facilitate the learning of new alphabets. We tested this on the Cyrillic alphabet bysystematically replacing certain Latin characters with their Cyrillic counterparts in sentences providing context. We captured eye-tracking data and directly assessed participants learning outcomes. Analysis of the eye-tracking data revealed no clear indications of learning, suggesting it may not be a suitable method for assessing learning in this context. In contrast, the results from direct assessments indicated that learning was achievable. Participants achieved the best performance when the initial letters were replaced.

A Appendix A

A.1 Predicting Testing Trial Outcomes from Eye-tracking Data

To further analyse what features could be important we predict the outcomes of the testing trials using the eye tracking data. To predict outcomes, we averaged all collected features for each letter and participant, then trained a logistic regression model with an L1 penalty for feature selection. We also evaluated a simple neural network with two linear layers, a ReLU activation function, and binary cross-entropy loss. The collected features included: total fixation duration, average fixation duration, number of fixations, average time to first fixation, minimum duration fixation, maximum duration fixation, regression path duration, rereading duration. Results are reported on a test set consisting of 20% of the data.

Logistic regression with an L1 penalty achieved 0.72% accuracy on the test set, equivalent to a majority classifier. Although the model was expected to perform feature selection, only one beta weight was non-zero, and this weight varied across runs. Using a neural network approach we also achieved 0.72 percent accuracy.

A.2 Trial metrics aggregated by category

Table 6 shows the aggregated metrics, grouped by letter category.

position	letter	sum	mean	std
Normal	B	20	1	0
	L	8	0.4	0.502
	D	12	0.666	0.485
Double	SH	18	0.9	0.307
	TS	10	0.5	0.512
	CH	7	0.35	0.489
Greek	G	14	0.7	0.47
	P	9	0.45	0.51
	F	15	0.75	0.444

Table 6: Aggregated metrics grouped by letter-category

Table 7 shows the ANOVA and Tukey's HSD for letter category distributions, making clear there is no significant difference between the groups.

ANOVA p-value		0.64		
Tukey's HSD results		greek vs double	greek vs normal	double vs normal
	p - adj	0.839	0.925	0.616

Table 7: ANOVA and Tukey's HSD for letter category distributions

We assessed learning success through two primary methods. First, we directly measured learning performance using the testing trials. Second, we examined changes in processing efficiency by analyzing eye-tracking data during repeated letter presentations. Specifically, we investigated whether revisits to and total viewing time of Cyrillic letters decreased with increasing exposure. Finally, we explored the predictive power of eye-tracking measures for learning outcomes using predictive modeling.

We hypothesized that maximum fixation duration and the number of fixations would decrease with increased familiarity. To test this, we compared these measures for the first five and last five presentations of each letter, using paired t-tests to determine statistical significance.

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